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Work submission

Only grade the work that is in the learner's or group's Git repository. Check that only the requested files are available in the git repository. If not, the evaluation stops here.

Yes

No

README.md

README.md file

Does the repository contain a README.md file at its root, and does it include all of the following?

- A "Description" section explaining the project's purpose and providing a brief overview.
- An "Instructions" section with details about how to run the project.
- An "Algorithm explanation" section describing the pathfinding approach and design decisions.
- Documentation of visual representation features and how they enhance user experience.
- Example input and expected output demonstrating the program's functionality.

Yes

No

Project Structure and Requirements

OOP

The project is completely object-oriented with proper class design and follows Python OOP best practices. Check for proper class separation, encapsulation, and inheritance where appropriate.

Yes

No

Type safe

The project is completely typesafe and passes mypy type checking without errors. Run 'mypy .' in the project directory to verify. If mypy is not installed, check that type hints are present throughout the codebase.

Yes

No

Graph implementation

The graph implementation is custom-built without using forbidden libraries (such as networkx, graphlib, etc.). The learner should be able to explain how it works.

Yes

No

Parser Implementation

Input files

Test the parser with valid input files. The parser correctly handles the new input file format including:

- Number of drones using "nb_drones: <number>" format
- Zone definitions with type prefixes (start_hub:, end_hub:, hub:)
- Connection definitions using "connection: <zone1>-<zone2>" format
- Optional metadata with defaults (zone=normal, max_drones=1)
- Connection metadata with max_link_capacity=1 default
- Comments starting with '#' are ignored

At least 4 out of 5 format requirements should work correctly.

Yes

No

Error handling

Test error handling in the parser. The parser correctly handles errors:

- Test with malformed files (missing drone count, invalid format)
- Test with invalid zone types (should reject with clear error messages)
- Test with missing start_hub or end_hub definitions
- Test with invalid capacity values (non-positive integers)
- Test with duplicate zone names or connections

At least 4 out of 5 error cases should be handled correctly with clear error messages.

Yes

No

Zone and Movements Mechanics

Zone occupancy rules

Test zone occupancy rules with capacity constraints:

- Zones respect max_drones capacity (default 1)
- Drones cannot enter zones that would exceed capacity
- Multiple drones can share start and end zones
- Capacity constraints are properly enforced during simulation

Create test cases to verify these rules are enforced.



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Yes

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Movement costs

Test movement costs based on zone types:

normal zones: 1 turn (default)

restricted zones: 2 turns

priority zones: 1 turn (but preferred in pathfinding)

blocked zones: inaccessible

Verify the simulation respects these movement costs in the output.

Yes

No

Connection capacity

Test connection capacity constraints:

Connections respect max_link_capacity defined in connection metadata (default 1)

Multiple drones can traverse high-capacity connections simultaneously

Connection capacity limits are enforced during movement

Verify capacity constraints work correctly in simulation.

Yes

No

Visual Representation

Visualization features

Test visual representation features:

The program provides a clear visual feedback (colored terminal output and/or graphical interface)

Colors specified in zone metadata are used for visualization

Visual representation enhances understanding of the simulation

The visual system clearly shows drone positions and movements

The implementation should demonstrate meaningful visual feedback.

Yes

No

Basic Functionality Tests

Simple scenarios

Test with simple scenarios:

- Test with single drone on a linear path

- Test with multiple drones using different paths

- Test with provided example maps from attachments/

- Verify output format: each line represents one turn, format "D<ID>-<zone>" or "D<ID>-<connection>"

- Verify stationary drones are omitted from output

At least 4 out of 5 basic tests should pass successfully.

Yes

No

Simulation ends correctly

Test that the simulation ends correctly when all drones reach the end zone.
The program should stop producing output once all drones have arrived.

Yes

No

Pathfinding Algorithm

Valid path

The pathfinding algorithm successfully finds valid paths from start to end zone for various test cases. Test with:

Simple linear maps

Maps with multiple possible paths

Maps with bottlenecks and capacity constraints

Maps using different zone types

The algorithm should find valid solutions for most test cases.

Yes

No

Conflict resolution

Test conflict resolution and capacity management:

Test scenarios where drones compete for limited capacity

Verify the algorithm handles multiple drones simultaneously

Check that capacity constraints prevent conflicts

Test turn-based movement with multi-turn zones (restricted)

The algorithm should demonstrate proper capacity-aware conflict resolution.

Yes

No



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Performance and Optimization

Efficiency

Test with more complex scenarios to evaluate performance:

- Test with higher drone counts (10+ drones)
- Test with complex map topologies and capacity constraints
- Verify the solution minimizes total simulation turns
- Test performance with large graphs

Note: This is not a failing requirement but should show reasonable performance.

 Yes

 No

Algorithm explanation

The learner can explain their algorithm complexity, design decisions, and trade-offs made in their implementation. They should be able to discuss their pathfinding approach, capacity management, and optimization strategies.

 Yes

 No

Performance Benchmarks

Easy maps

Test performance against reference benchmarks using provided maps:

Does the algorithm solve the **Easy maps** in less than 10 turns on average?

Reference targets (for information only):

- Linear path (2 drones): ≤ 6 turns
- Simple fork (4 drones): ≤ 8 turns
- Basic capacity (4 drones): ≤ 6 turns These reference values are provided as optimization goals to help learners evaluate and tune their algorithms.

 Yes

 No

Medium maps

Test performance against reference benchmarks using provided maps:

Does the algorithm solve the **Medium maps** in 10–30 turns on average?

Reference targets (for information only):

- Dead end trap (5 drones): ≤ 12 turns
- Circular loop (6 drones): ≤ 15 turns
- Priority puzzle (5 drones): ≤ 12 turns

These reference values are provided as optimization goals to help learners evaluate and tune their algorithms.

 Yes

 No

Hard maps

Test performance against reference benchmarks using provided maps:

Does the algorithm solve the **Hard maps** in less than 60 turns on average?

Reference targets (for information only):

- Maze nightmare (8 drones): ≤ 30 turns
- Capacity hell (12 drones): ≤ 35 turns
- Ultimate challenge (15 drones): ≤ 45 turns These reference values are provided as optimization goals to help learners evaluate and tune their algorithms.

 Yes

 No

Edge Cases and Error Handling

Edge cases

Test edge cases:

- Single drone scenarios
- Maps with bottlenecks or limited capacity
- Disconnected graphs (should handle gracefully)
- Invalid connections between zones
- Zero or very high capacity values

At least 4 out of 5 edge cases should be handled correctly.

 Yes

 No

Error handling

The program provides appropriate error handling for invalid inputs:

- Malformed files with new format requirements
- Missing start_hub or end_hub zones
- Invalid capacity values
- Disconnected graphs
- Invalid zone types or connections

Error messages should be clear and informative.

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Code Quality and Documentation

Code quality

Code quality assessment:

- Code is well-structured and readable
- Proper use of object-oriented principles
- Appropriate comments and documentation
- Consistent coding style
- Visual representation code is well-integrated

Note: This is not a failing requirement but contributes to overall quality.

Yes

No

Quick Live Coding modifications

Live coding

Please ask the reviewee to add a simple modification to their drone simulation program. Ask them to add a "--capacity-info" flag that displays capacity information during simulation, such as "Zone X: Y/Z drones, Connection A-B: Y/Z capacity used" for each turn.

For example, "./main.py --capacity-info map.txt" should output the normal simulation plus capacity usage information. The reviewee should be able to locate the relevant parsing and output code, make the necessary modifications, and demonstrate that it works with a test case. The entire task, including the demonstration, should take no more than 10 minutes. Was this procedure followed and did everything work correctly?

Yes

No

Bonus Features

Exceptional Performance

The Performance 'perfectly' meets the performance reference targets for all provided maps. 'Perfectly' meaning the algorithm meets exactly or outperforms the number of turns specified in the reference targets for each map category.

Targets are:

Easy maps:

- Linear path (2 drones): ≤ 6 turns
- Simple fork (4 drones): ≤ 8 turns
- Basic capacity (4 drones): ≤ 6 turns

Medium maps:

- Dead end trap (5 drones): ≤ 12 turns
- Circular loop (6 drones): ≤ 15 turns
- Priority puzzle (5 drones): ≤ 12 turns

Hard maps:

- Maze nightmare (8 drones): ≤ 30 turns
- Capacity hell (12 drones): ≤ 35 turns
- Ultimate challenge (15 drones): ≤ 45 turns

Yes

No

Challenger Map

The Impossible Dream map is solved and beats the reference record of 45 turns.

Yes

No

Ratings

Don't forget to check the flag corresponding to the defense

Ok

★ Outstanding project

Empty work

Incomplete work

Invalid compilation

Norme

Cheat

Crash

Incomplete group

Concerning situation

Leaks

Forbidden function

Can't support / explain code

Conclusion

Leave a comment on this evaluation (2048 chars max)

Finish evaluation



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